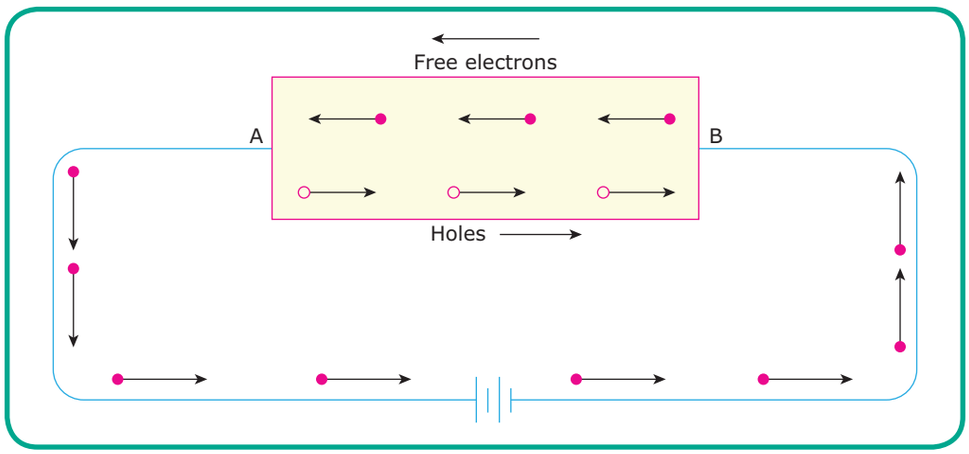


Figure 3.21 **Electron-hole current**

holes. Therefore, the total current inside the semiconductors is the sum of currents due to free electrons and holes.

It may be noted that current in the external wires is fully electronic i.e., by electrons. Then what about holes? Referring to the Figure 3.21, holes being positively charged and move towards the negative potential of supply. As the holes reach the negative terminal B, electrons enter the semiconductor crystal near the terminal and combine with holes, thus cancelling each other. At the same time, the loosely held electrons near by the positive terminal *A* are attracted away from their atoms into the positive terminal. This creates new holes near the positive terminal which again drift towards the negative terminal.

3.7.1 Extrinsic Semiconductors

The pure semiconductor must be altered so as to significantly increase its conductive properties. This is achieved by adding a small amount of suitable impurity to semiconductors. It is then called impurity or extrinsic semiconductors.

The process of adding impurities to a semiconductor is known as doping.

Generally, for 108 atoms of semiconductor, one impurity atom is added.

The purpose of adding impurity is to increase either the number of free electrons or holes in the semiconductor crystal. If the pentavalent impurity (having 5 valence electrons) is added to the semiconductor, a large number of electrons are produced in the semiconductor. On the other hand, addition of trivalent impurity (Having 3 valence electron) to semiconductor generates large number of holes.

Depending upon the type of impurity added extrinsic semiconductors are classified into:

1. N-type Semiconductor
2. P-type Semiconductor

3.8 N-TYPE SEMICONDUCTOR

When a small amount of pentavalent element is added to pure semiconductor, it is known as N-type semiconductors.

The addition of pentavalent impurity provides a large number of free electrons in the semiconductor crystal. Typical

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